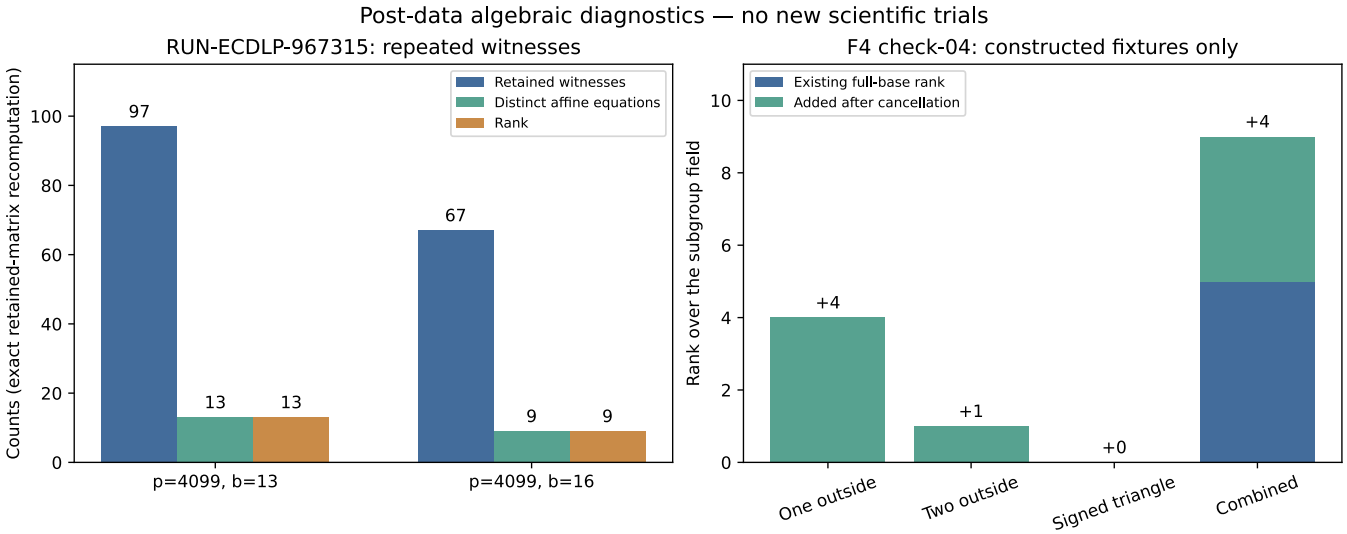
Pursuing four prime-field factor-base leads

28 October 2026 · continuing plan and exploratory retained-data analysis

No new scientific trial started. This report separates recomputed historical matrices, proposed next experiments and an observed runtime refusal. The full plan, exact inputs, controls and result-dependent successors are in PLAN.md.



What the retained data show

The two p=4099 Prym matrices contain 97/67 witnesses but 13/9 distinct affine equations, rank 13/15 and 9/15, and nullity 2/6. No individual log column is identified. Each matrix identifies one two-column weighted combination chosen after observing it. Those combinations are constructed controls, not natural unknown-scalar targets. Both fixtures contain the public x=0 generator as their first atom, so testing zero completion requires no larger base.

The retained combined F4 fixture has existing full-base rank 5. Eliminating its outside columns adds 4 independent ranks to reach 9. The separate one-outside/two-outside fixtures add 4/1, and the signed triangle adds 0. These fixtures were constructed. No natural collision or throughput estimate can be inferred. All input hashes and equation certificates are retained.

First experimental wave

B1, isolated zero completion: same two curves, 16 atoms, 64-source order and cubic law; fresh historical and completed-zero arms. The full paired workload is 9,175,040 scheduled branches, derived before execution. Score distinct equations and old-kernel syndromes before rank, retaining every failure.

C1, Dickson depth 4: exact existing F1 inputs and all multiplicity strata, maximum operator dimension 560, complete signed/state oracle, nonseparable and corrupted-certificate controls. Record costs by stage; small degree and a completed toy case are not complexity evidence.

A1, useful partial rows: consume complete naturally generated signed partial rows and compare cancellation with the dense oracle on identical rows. Charge outside vertices, zero combinations, duplicates and elimination. No random-graph collision assumption is justified automatically.

D1/D2, natural targets: freeze public hash-derived targets independently of the matrix; first test row identifiability, then combinations in an augmented system retaining the target coefficient. Full base rank is not required. Eight proposed targets have not been generated. Existing full-rank E-stage gates remain unchanged; this is a separate target-specific study.

Actual execution obstacle and repair

The canonical checkpoint first failed on GNU/Linux stat portability. The metadata repair passes sixteen checkpoint/relay tests. The repaired gate then exited 1 because no supported token/AWS transport is configured. No upload or scientific process started. Both attempts and the missing sparse-fixture test failure are preserved. The linked problem record names exact exit conditions.

All four directions remain open with concrete successors in PLAN.md; none is retired by a finite null observation. No hypothesis, bound or scientific status changes. PR2664 is merged; no unattended or scheduled experiment is running.